PRACTICAL 8

MySQL Schema Design, ER Model, PDO Connectivity and Prepared Statements

Problem Definition: Design the database for StudentHub with students, events and registrations. Create a secure PHP database connection using MySQL and PDO.

Output: XAMPP Services Running

Service	State	Ports / evidence
Apache Web Server	RUNNING	Ports 80 and 443
MySQL Database	RUNNING	Port 3306; XAMPP panel reports active

StudentHub Database Structure

Table	Primary key	Relationships / constraints	Rows
courses	course_id	Unique course_name	4
students	id	Unique email; FK course_id -> courses	4
events	event_id	Unique title + event_date	4
registrations	reg_id	FK student_id + event_id; unique pair	4

Database: studenthub_practical8 | Engine: InnoDB | Charset: utf8mb4_unicode_ci

The normalized courses table avoids repeating course names in student rows. Registrations is the junction table for the many-to-many student/event relationship.

Sample Event Registration Records

Registration	Student ID	Event ID	Registered on
1	1	1	2026-10-28 12:09
2	1	4	2026-10-28 12:09
3	2	4	2026-10-28 12:09
4	3	2	2026-10-28 12:09

Sample Event Records

ID	Title	Date	Location	Category
1	Career Guidance Session	2026-10-12	Seminar Hall	Career
2	Sports Meet	2026-10-15	College Ground	Sports
3	Photography Workshop	2026-10-20	Media Room	Workshop
4	Coding Hackathon	2026-10-22	Innovation Lab	Technology

The unique constraint on (student_id, event_id) prevents the same student from registering for the same event twice.

Sample Student Records

ID	Name	Email	Course	Year	Gender
1	Jash Tanna	jash.tanna@studenthub.edu	Information Technology	2	Male
2	Aarav Shah	aarav.shah@studenthub.edu	Computer Science	3	Male
3	Diya Patel	diya.patel@studenthub.edu	Information Technology	1	Female

Successful PDO Database Connection

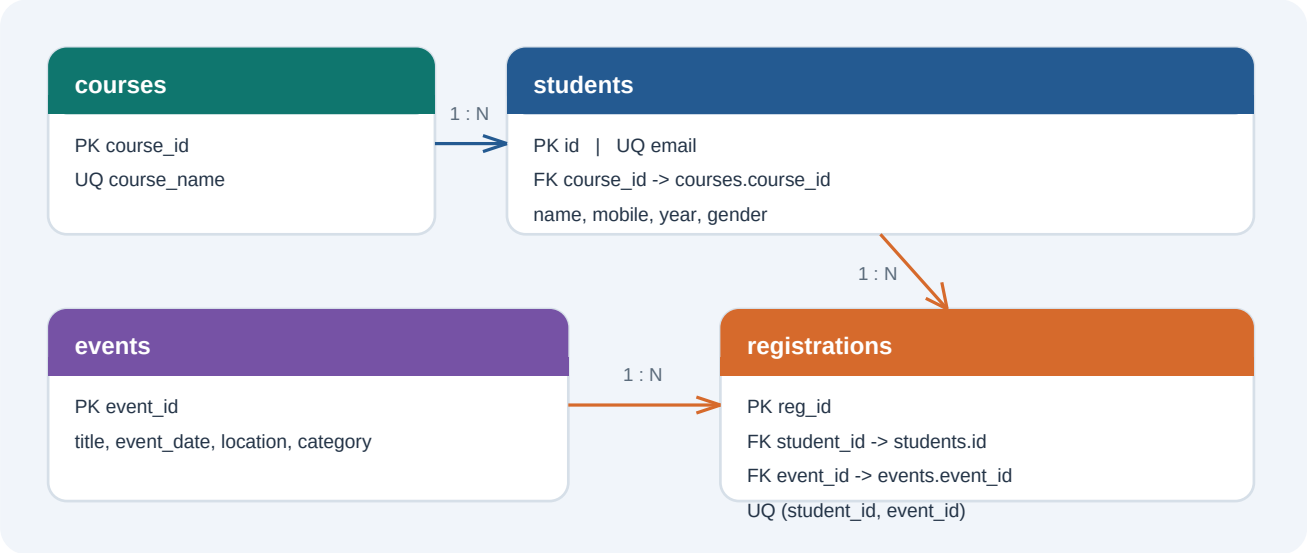
PDO connected to the StudentHub database successfully.

Database: **studenthub_practical8** Server: **10.4.32-MariaDB**

Connection verified through php/db_test.php while XAMPP Apache and MySQL were running.

Connection details and passwords are not displayed. PDO uses exception mode and native prepared statements.

StudentHub Entity Relationship Diagram (ERD)



Question & Answer

1. Is the schema normalized properly?

Answer: Yes. Course names are stored once in courses. Student, event and registration details are in separate tables, reducing duplication.

2. Are primary keys and foreign keys correctly defined?

Answer: Yes. Each table has a primary key. students.course_id references courses, while registrations.student_id and event_id reference students and events.

3. Is the database connection handled securely?

Answer: Yes. PHP uses PDO with exception handling and prepared statements. Errors are logged server-side; users see a generic connection message.

Conclusion / Learning Outcome

StudentHub stores related student, event and registration data in a normalized MySQL database. Primary and foreign keys protect relationships, while PDO prepared statements support secure database-backed pages.